

Software Requirements Specification (SRS) ThesisSync

A Web-Based Thesis Collaboration Platform for Students, Supervisors, and Co-supervisors

Document Control

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Glossary

Term	Definition
API	Application Programming Interface
CRUD	Create, Read, Update, Delete
CSE	Computer Science and Engineering
DR	Data Requirement
ERD	Entity Relationship Diagram
FR	Functional Requirement
HTTPS	Hypertext Transfer Protocol Secure
IR	Interface Requirement
JWT	JSON Web Token
KiU	Kishoreganj University
NFR	Non-Functional Requirement
ORM	Object-Relational Mapping
RBAC	Role-Based Access Control
REST	Representational State Transfer
SR	Security Requirement
SRS	Software Requirements Specification
TLS	Transport Layer Security
UI/UX	User Interface / User Experience
WCAG	Web Content Accessibility Guidelines

1. Introduction

1.1. Purpose

This Software Requirements Specification (SRS) document provides a comprehensive description of ThesisSync, a web-based thesis collaboration platform to be developed for undergraduate thesis teams, supervisors, and co-supervisors at the Department of Computer Science and Engineering, Kishoreganj University. The document details the functional and non-functional requirements of the platform, which aims to centralize thesis-related communication, task tracking, document management, and supervisor feedback within a single workspace.

This SRS serves as the foundation for the subsequent system design and development phases of CSE4106, Software Development Lab-1, ensuring that all stakeholders - the project team and the course instructor - share a clear understanding of what the system will do and how it will operate.

1.2. Document Conventions

This document follows the conventions below:

Term	Description
SHALL	Refers to a mandatory requirement that must be fulfilled in the initial implementation of the platform.
SHOULD	Indicates a requirement recommended for the current phase, or planned for a near-future iteration, that the developers should take into consideration.
MAY	Refers to an optional requirement, or one anticipated for a future phase beyond the current course deliverable, kept in mind for scalability.
TBD	To Be Determined - indicates information not yet available, to be finalized during the design phase.

Requirements are categorized as follows:

Prefix	Description
FR-XX	Functional Requirements
NFR-XX	Non-Functional Requirements
IR-XX	Interface Requirements

Prefix	Description
DR-XX	Data Requirements

1.3. Intended Audience

This document is intended for the following stakeholders:

Stakeholder	Role
Course Instructor	Md. Sabab Zulfiker, Lecturer - reviews and approves the requirements and evaluates the deliverable.
Project Team	Md. Ehasan Mahmud and Meherab Hossen Opi - design, implement, and test the system.
Thesis Students	End users who register, manage tasks, maintain journals, and submit documents.
Supervisors / Co-supervisors	End users who review submissions, provide feedback, and approve milestones.
Department Administrators	Maintain user accounts, thesis groups, and platform configuration.

1.4. Project Scope

ThesisSync centralizes thesis collaboration for undergraduate thesis teams, providing a single workspace for two student members, a supervisor, and a co-supervisor. It replaces the current reliance on messaging apps, email, and scattered cloud storage with a dedicated academic collaboration environment.

In Scope

The platform will include the following key components:

- User Management System: registration and profile management for students, supervisors, co-supervisors, and administrators
- Role-Based Access Control: distinct access levels for each user role
- Project Dashboard: a consolidated view of tasks, deadlines, files, and feedback
- Task Management: creation, assignment, and status tracking of thesis-related tasks
- Progress Journal: dated log entries documenting research progress
- File Repository: structured document submission with versioning

- Review Comments: structured supervisor feedback tied to specific submissions
- Milestone Tracking: defined thesis milestones with approval workflow
- Meeting Scheduler: meeting proposals, calendar view, and recorded minutes
- Notification System: email and in-app notifications for key events

Out of Scope

The following items are explicitly excluded from this phase of the project:

- AI-based features (e.g., automated writing suggestions, smart recommendations)
- Plagiarism detection
- Real-time collaborative document editing
- Video conferencing
- A native mobile application
- Data migration from any pre-existing system (ThesisSync is a greenfield deployment)

1.5. References

This SRS is based on the following documents and sources:

- ThesisSync Project Proposal, CSE4106: Software Development Lab-1, submitted June 6, 2026
- Course guidelines and evaluation criteria for CSE4106, Department of CSE, Kishoreganj University
- Discussions and requirement clarification with the course instructor, Md. Sabab Zulfiker
- General best practices for academic collaboration and project-management platforms

2. Overall Description

2.1. Product Perspective

ThesisSync is a new, standalone web-based system developed specifically for the thesis workflow within the Department of Computer Science and Engineering. It is not part of a larger existing system, though it is designed with a modular architecture that would allow future integration with departmental systems (e.g., a student information system), should the department choose to expand its use beyond this course.

The platform interacts primarily with its own database and file storage, and with an email service provider for outbound notifications. No integration with external job sites, government databases, or third-party AI services is required.

2.2. Product Functions

ThesisSync will provide the following major functions:

1) User Registration and Profile Management

- Student registration and profile creation
- Supervisor and co-supervisor registration and profile management
- Administrator account and thesis-group management

2) Project Dashboard

- Consolidated view of tasks, deadlines, files, and recent activity
- Supervisor view across all assigned thesis groups

3) Task Management

- Task creation, assignment, and status tracking
- Task-level comments and due-date reminders

4) Progress Journal

- Dated journal entries with optional attachments
- Supervisor visibility and commenting on journal entries

5) File Repository and Document Submission

- Categorized document upload with version history
- Structured review comments tied to submissions
- Milestone definition, tracking, and approval

6) Meeting Scheduler

- Meeting proposals, responses, and calendar view
- Recorded meeting minutes and history

7) Notification System

- Email notifications for key events
- In-app notification center

2.3. User Classes, Characteristics, and Needs

The platform will serve the following user classes:

1) Thesis Students

- Undergraduate CSE students in groups of two, working on a shared thesis project
- Comfortable with standard web applications; varying availability and internet access
- Primary needs: coordinating tasks, tracking progress, submitting documents, and receiving timely supervisor feedback

2) Supervisors and Co-supervisors

- Faculty members responsible for guiding one or more thesis groups
- Need an efficient way to monitor multiple groups without switching between disconnected tools
- Primary needs: reviewing submissions, providing structured feedback, approving milestones, scheduling meetings

3) Administrators

- Departmental staff or course coordinators responsible for platform oversight
- Primary needs: managing accounts and thesis-group assignments, monitoring system-wide activity

2.4. Operating Environment

1) Technical Environment

- A web-based application accessible via standard browsers
- Responsive design for access via smartphones and tablets
- Cloud-hosted Linux server, per the project's deployment plan

2) Hardware Environment

- Standard cloud server infrastructure sized for a single-department, course-scale user base
- Adequate storage for user data, documents, and journal attachments

3) Software Environment

- Modern web technologies and frameworks for frontend and backend development

- A relational database management system (e.g., MySQL)
- Standard web-security libraries for authentication and data protection

4) User Environment

- Desktop computers, laptops, tablets, and smartphones
- Modern browsers (Chrome, Firefox, Safari, Edge)
- Primary interface language: English

2.5. Design and Implementation Constraints

1) Technical Constraints

- Must be accessible via standard web browsers without special plugins
- Must be implementable and deployable within the CSE4106 course timeline (16 weeks)
- Must be maintainable by the two-person project team with the technologies taught in the course

2) Regulatory Constraints

- Must respect the university's data handling and academic integrity policies
- Must apply reasonable data-protection practices for user credentials and personal information

3) Business/Academic Constraints

- Must be implemented in phases, with core functionality (authentication, tasks, file repository) prioritized over secondary features (analytics, advanced reporting)
- Must operate within the estimated project budget of BDT 8,000

2.6. User Documentation

- Online help / FAQ section covering common tasks for each role
- Short user guides: Student Guide, Supervisor Guide, Administrator Guide
- A brief system walkthrough (screenshots or a short recorded demo) for course evaluation and onboarding

2.7. Assumptions and Dependencies

Assumptions

- The course instructor and any pilot users will be available for requirement validation and feedback during the project timeline.
- The project team will have continued access to a cloud hosting environment for development and deployment.
- The initial release will be used by a limited number of pilot thesis groups rather than the entire department.

Dependencies

- Availability of a reliable email service (SMTP) for account verification and notifications.
- Availability of the two-person development team's time alongside other coursework.
- Timely feedback from the course instructor during system design and testing.

3. System Features and Functional Requirements

This section details the functional requirements of ThesisSync, organized by major system feature. Each requirement is identified with a unique ID (FR-XX) for traceability.

3.1. User Management and Authentication

3.1.1. Student (Thesis Team Member) Registration and Profile Management

ID	Requirement
FR-01	The system SHALL provide a self-registration process for thesis team members using their university student ID and institutional email address.
FR-02	The system SHALL require account activation via a verification link sent to the student's registered email.
FR-03	The system SHALL collect the following information from students through a staged onboarding process: Level 1 — full name, student ID, email, mobile number, password, department, batch/session; Level 2 — working thesis title, research area/topic tags, group number, profile photo.
FR-04	The system SHALL allow students to view and update their profile information at any time.
FR-05	The system SHALL associate each student account with exactly one active thesis group at a time.

ID	Requirement
FR-06	The system SHALL display, on each student's profile, the names and roles of their thesis group members, supervisor, and co-supervisor.
FR-07	The system SHALL allow students to set a profile visibility preference, restricted by default to their group, supervisor, and co-supervisor.

3.1.2. Supervisor and Co-Supervisor Registration and Profile Management

ID	Requirement
FR-08	The system SHALL provide a registration process for supervisors and co-supervisors using their institutional email and faculty ID.
FR-09	The system SHALL require account activation via email verification for faculty accounts.
FR-10	The system SHALL allow supervisors to view a list of all thesis groups currently assigned to them.
FR-11	The system SHALL allow a faculty member to be assigned as either the primary supervisor or co-supervisor of a given thesis group.
FR-12	The system SHALL allow supervisors to maintain a profile listing their research interests, designation, and contact information.
FR-13	The system SHALL allow supervisors to switch between multiple assigned thesis groups from a single account.

3.1.3. Administrator User Management

ID	Requirement
FR-14	The system SHALL provide an administrator interface for creating, editing, and deactivating user accounts (students, supervisors, co-supervisors).
FR-15	The system SHALL allow administrators to create thesis groups and assign students, a supervisor, and a co-supervisor to each group.
FR-16	The system SHALL allow administrators to reset user passwords and assist with account recovery.
FR-17	The system SHALL maintain an audit log of all administrative actions for accountability.

ID	Requirement
FR-18	The system SHALL allow administrators to view system-wide statistics, including the number of active groups, submissions, and overdue milestones.

3.1.4. Authentication and Authorization

ID	Requirement
FR-19	The system SHALL implement authentication using email and password, with passwords stored using a one-way hashing algorithm.
FR-20	The system SHALL enforce a minimum password strength policy (minimum length and character variety).
FR-21	The system SHALL implement role-based access control (RBAC) with at least four roles: Student, Supervisor, Co-supervisor, and Administrator.
FR-22	The system SHALL provide a "forgot password" flow using a time-limited email reset link.
FR-23	The system SHALL automatically end inactive sessions after a configurable timeout period.

3.2. Project Dashboard

ID	Requirement
FR-24	The system SHALL provide each thesis group with a shared project dashboard summarizing tasks, upcoming deadlines, recent files, and recent comments.
FR-25	The system SHALL display an overall progress indicator (percentage of milestones completed) on the group dashboard.
FR-26	The system SHALL display, for supervisors, a consolidated dashboard listing all assigned groups with their current status (on track, at risk, overdue).
FR-27	The system SHALL display upcoming meeting dates on the dashboard.
FR-28	The system SHALL visually highlight overdue tasks and unread supervisor feedback on the dashboard.

3.3. Task Management

ID	Requirement
FR-29	The system SHALL allow group members to create, edit, and delete tasks within their thesis project.
FR-30	The system SHALL allow a task to be assigned to one or more group members with a due date.
FR-31	The system SHALL support the following task statuses: To Do, In Progress, Completed, Blocked.
FR-32	The system SHALL allow supervisors and co-supervisors to view (read-only) the task list of their assigned groups.
FR-33	The system SHALL allow group members to add comments to individual tasks.
FR-34	The system SHALL send a reminder notification when a task's due date is approaching or has passed.

3.4. Progress Journal

ID	Requirement
FR-35	The system SHALL allow each thesis group to maintain a shared progress journal with dated entries.
FR-36	The system SHALL allow journal entries to include a text description and optional attached files or links.
FR-37	The system SHALL allow supervisors and co-supervisors to view and comment on journal entries of their assigned groups.
FR-38	The system SHALL display journal entries in reverse-chronological order, filterable by date range and author.
FR-39	The system SHALL allow students to edit journal entries within a limited time window after creation, retaining an edit history.

3.5. File Repository and Document Submission

3.5.1. Document Upload and Versioning

ID	Requirement
FR-40	The system SHALL allow group members to upload thesis-related documents in common formats (PDF, DOCX, PPTX).
FR-41	The system SHALL organize uploaded documents into categories such as Proposal, Progress Report, Draft Chapters, Final Thesis, and Presentation.
FR-42	The system SHALL retain previous versions of a document whenever a new version is uploaded, and SHALL allow authorized users to view or restore prior versions.
FR-43	The system SHALL display file metadata - uploader, upload date/time, file size, and version number - for every document.
FR-44	The system SHALL enforce a maximum file size limit per upload, configurable by administrators.
FR-45	The system SHALL allow group members to organize documents into folders within their group's file repository.

3.5.2. Review Comments and Feedback

ID	Requirement
FR-46	The system SHALL allow supervisors and co-supervisors to leave comments on a specific submitted document.
FR-47	The system SHALL tie comments to a specific document version.
FR-48	The system SHALL notify group members when new feedback is added to one of their submissions.
FR-49	The system SHALL allow students to mark a feedback item as "addressed" once resolved.
FR-50	The system SHALL maintain a complete feedback history for each document, viewable by all group members and supervisors.

3.5.3. Milestone Tracking and Approval

ID	Requirement
FR-51	The system SHALL allow administrators or supervisors to define thesis milestones (e.g., Proposal Defense, Mid-Term Review, Final Defense)

ID	Requirement
	with target dates.
FR-52	The system SHALL allow supervisors to mark a milestone as Approved, Needs Revision, or Rejected, with accompanying comments.
FR-53	The system SHALL automatically update a group's dashboard progress indicator when a milestone status changes.
FR-54	The system SHALL notify all group members when a milestone status is updated.
FR-55	The system SHALL maintain a timeline view showing all milestones and their current status for each thesis group.

3.6. Meeting Scheduler

ID	Requirement
FR-56	The system SHALL allow supervisors or students to propose a meeting with a date, time, and agenda.
FR-57	The system SHALL allow invited participants to accept, decline, or propose an alternate time for a meeting request.
FR-58	The system SHALL maintain a calendar view of upcoming and past meetings for each thesis group.
FR-59	The system SHALL allow meeting organizers to record minutes or notes after a meeting concludes, visible to all group members.
FR-60	The system SHALL send reminder notifications ahead of scheduled meetings (e.g., 24 hours and 1 hour prior).
FR-61	The system SHALL maintain a searchable history of past meetings and their recorded notes for each group.

3.7. Notification System

3.7.1. Email Notifications

ID	Requirement
FR-62	The system SHALL send email notifications for key events: new task assignment, new comment/feedback, milestone status change, and

ID	Requirement
	meeting scheduling.
FR-63	The system SHALL allow users to configure which event types trigger an email notification.
FR-64	The system SHALL log all outgoing email notifications for troubleshooting purposes.

3.7.2. In-App Notifications

ID	Requirement
FR-65	The system SHALL provide an in-app notification center displaying unread and past notifications.
FR-66	The system SHALL display a notification badge/count on the dashboard when unread notifications exist.
FR-67	The system SHALL allow users to mark notifications as read, or clear them individually or in bulk.

3.8. Role Management and Access Control

ID	Requirement
FR-68	The system SHALL restrict visibility of a thesis group's data (tasks, journal, files, comments) to that group's members, its supervisor, its co-supervisor, and administrators.
FR-69	The system SHALL prevent students from accessing or viewing data belonging to thesis groups they are not a member of.
FR-70	The system SHALL allow administrators to grant temporary read-only access to a thesis group's data for external evaluators (e.g., defense committee members), when required.

4. External Interface Requirements

4.1. User Interfaces

The system shall provide user interfaces for the following user types:

1) Students

- Registration and profile management interface
- Task board and progress journal interface
- File repository and submission interface
- Meeting and notification interface

2) Supervisors and Co-supervisors

- Multi-group dashboard interface
- Document review and feedback interface
- Milestone approval interface
- Meeting scheduling interface

3) Administrators

- User and thesis-group management interface
- System configuration interface
- System-wide activity and statistics interface

All user interfaces shall adhere to the following requirements:

- Responsive design for desktop, tablet, and mobile devices
- Consistent navigation and design patterns across all pages
- Clear, role-appropriate dashboards showing the most relevant information first
- Adherence to WCAG 2.1 Level AA guidelines where practical

4.2. Hardware Interfaces

- Application and database servers hosted on a cloud-based Linux environment
- Client devices: desktop computers, laptops, tablets, and smartphones
- No specialized or proprietary hardware is required

4.3. Software Interfaces

- Relational database management system (e.g., MySQL) for persistent data storage

- SMTP-based email service provider for account verification and notification delivery
- Standard web application framework for frontend and backend development
- Version control system (Git) for source-code management

4.4. Communications Interfaces

- HTTPS for all web traffic between client browsers and the server
- SMTP for outbound email notifications
- A RESTful API using JSON as the primary data-exchange format between frontend and backend

All communications shall be secured using appropriate encryption and authentication mechanisms.

5. Non-Functional Requirements

This section specifies the non-functional requirements for ThesisSync - the quality attributes and constraints the system must satisfy, scaled appropriately for a course-scale academic collaboration platform.

5.1. Performance Requirements

ID	Requirement
NFR-01	The system SHALL provide page load times of less than 3 seconds for standard operations under normal load conditions.
NFR-02	The system SHALL complete file uploads and downloads (up to the configured size limit) within 10 seconds on a standard broadband connection.
NFR-03	The system SHALL load the group dashboard summary within 2 seconds.
NFR-04	The system SHALL support at least 50 concurrent users without noticeable performance degradation, consistent with a single-department, course-scale deployment.
NFR-05	The system SHALL return search and filter results for journal entries and files within 2 seconds.

5.2. Security Requirements

ID	Requirement
NFR-06	The system SHALL encrypt all data in transit using HTTPS/TLS.
NFR-07	The system SHALL store passwords using a strong one-way hashing algorithm (e.g., bcrypt).
NFR-08	The system SHALL enforce role-based access control on every API endpoint.
NFR-09	The system SHALL expire session tokens after a configurable period of inactivity.
NFR-10	The system SHALL maintain access logs for all login attempts, successful and failed.
NFR-11	The system SHALL implement standard protections against common web vulnerabilities, including SQL injection, cross-site scripting (XSS), and cross-site request forgery (CSRF).
NFR-12	The system SHALL limit repeated failed login attempts through rate limiting or temporary account lockout.

5.3. Reliability and Availability

ID	Requirement
NFR-13	The system SHALL maintain at least 99% uptime during the academic semester, excluding scheduled maintenance.
NFR-14	The system SHALL perform automated daily backups of the database and file repository.
NFR-15	The system SHALL retain backups for at least 30 days, stored separately from the primary server.
NFR-16	The system SHALL display graceful, user-friendly error messages without exposing internal system details such as stack traces.
NFR-17	The system SHALL recover automatically from transient failures, such as temporary database connection loss, through retry mechanisms.

5.4. Usability and Accessibility

ID	Requirement
NFR-18	The system SHALL provide a consistent and intuitive user interface across all pages.
NFR-19	The system SHALL implement responsive design to support desktop, tablet, and mobile browsers.
NFR-20	The system SHALL follow WCAG 2.1 Level AA guidelines where practical, including sufficient color contrast, keyboard navigation, and text alternatives for non-text content.
NFR-21	The system SHALL provide clear onboarding guidance for first-time users.
NFR-22	The system's primary interface language SHALL be English; interface text SHALL be organized to allow future translation (e.g., to Bengali).

5.5. Maintainability and Portability

ID	Requirement
NFR-23	The system SHALL be built with a modular architecture separating frontend, backend, and database layers.
NFR-24	The system SHALL follow consistent coding standards and include inline documentation.
NFR-25	The system SHALL use Git version control for all source code, with a documented branching strategy.
NFR-26	The system SHALL be deployable to a standard Linux-based cloud host without dependence on a single proprietary vendor.
NFR-27	The system SHALL externalize configuration values (e.g., file size limits, email/SMTTP settings) rather than hard-coding them.

5.6. Legal and Compliance Requirements

ID	Requirement
NFR-28	The system SHALL comply with Kishoreganj University's applicable data handling and academic integrity policies.
NFR-29	The system SHALL collect and store only the data necessary for thesis

ID	Requirement
	collaboration (data minimization).
NFR-30	The system SHALL allow users to request a review or deletion of their account data after project or course completion, subject to university record-keeping requirements.
NFR-31	The system SHALL properly license and attribute all third-party libraries and frameworks used in its development.

5.7. Operational Requirements

ID	Requirement
NFR-32	The system SHALL implement basic logging of system errors and key events for troubleshooting.
NFR-33	The system SHALL provide a simple health-check endpoint for monitoring uptime.
NFR-34	The system SHALL be accompanied by documented backup and restore procedures for the responsible system administrator.
NFR-35	The system SHALL include setup and deployment documentation sufficient for a future development team to redeploy or hand off the system.

6. Other Requirements

6.1. Data Migration

Not applicable. ThesisSync is a new (greenfield) system with no legacy data to migrate. All user accounts, thesis groups, and content will be created directly within the platform from initial deployment.

6.2. Internationalization Requirements

- The primary interface language for the initial release is English.
- Interface text SHALL be organized separately from application logic to allow future translation, such as to Bengali, without a redesign.
- Dates and times SHALL be displayed in a locale-appropriate format, using the Asia/Dhaka timezone by default.

6.3. Training Requirements

- A short onboarding guide will be provided for students and supervisors, covering registration, task management, and document submission.
- A brief walkthrough or demo will be delivered during the course presentation and evaluation.
- No formal, multi-session training program is required given the platform's limited pilot user base for this course deliverable.

6.4. Appendix A: Analysis Models

This appendix presents the core analysis models produced during the System Design phase, illustrating the system's boundaries, data flows, key process, data model, and use cases.

A.1. Context Diagram

The context diagram shows ThesisSync as a single system and its interactions with external entities: students, supervisors, co-supervisors, administrators, and the email service used for outbound notifications.

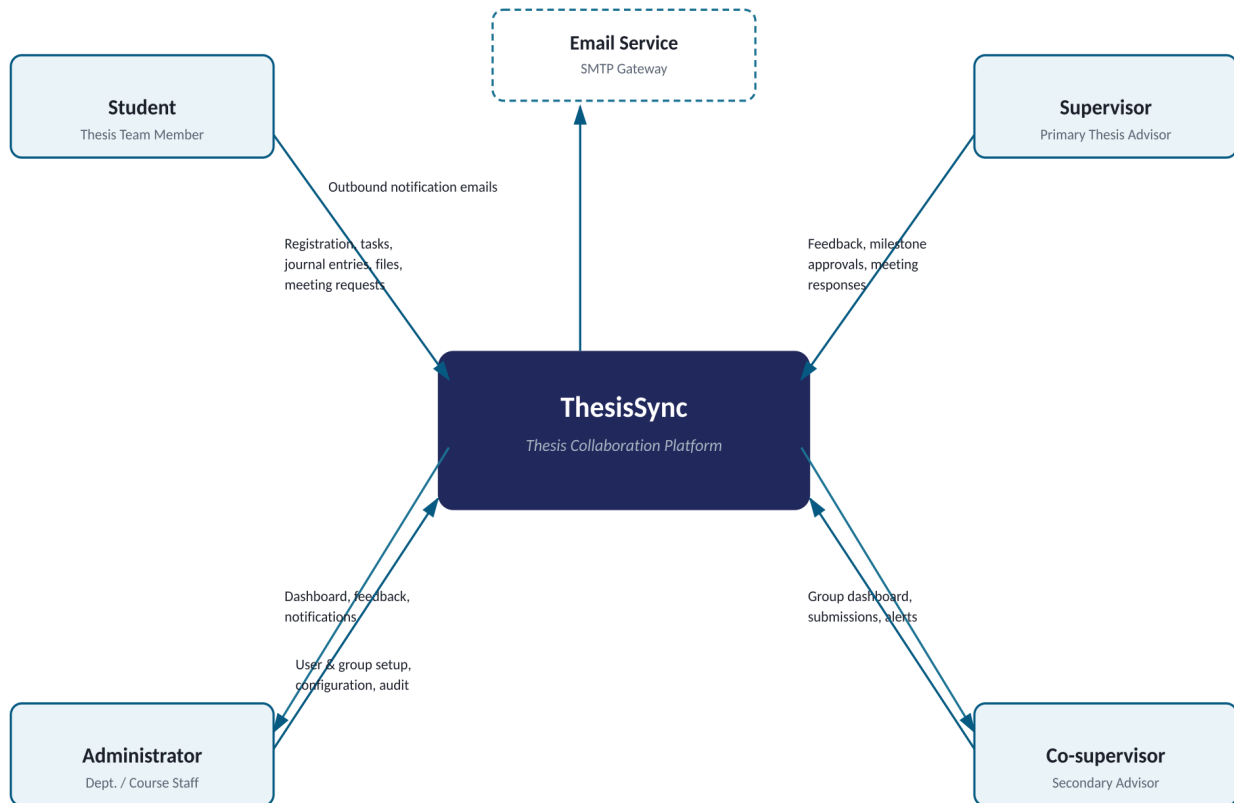


Figure A.1 - ThesisSync Context Diagram (Level 0)

A.2. Data Flow Diagram (Level 1)

The Level 1 data flow diagram decomposes ThesisSync into its five major processes - user and group management, task and journal management, document and milestone management, meeting management, and notification management - and shows how data moves between external entities, processes, and data stores.

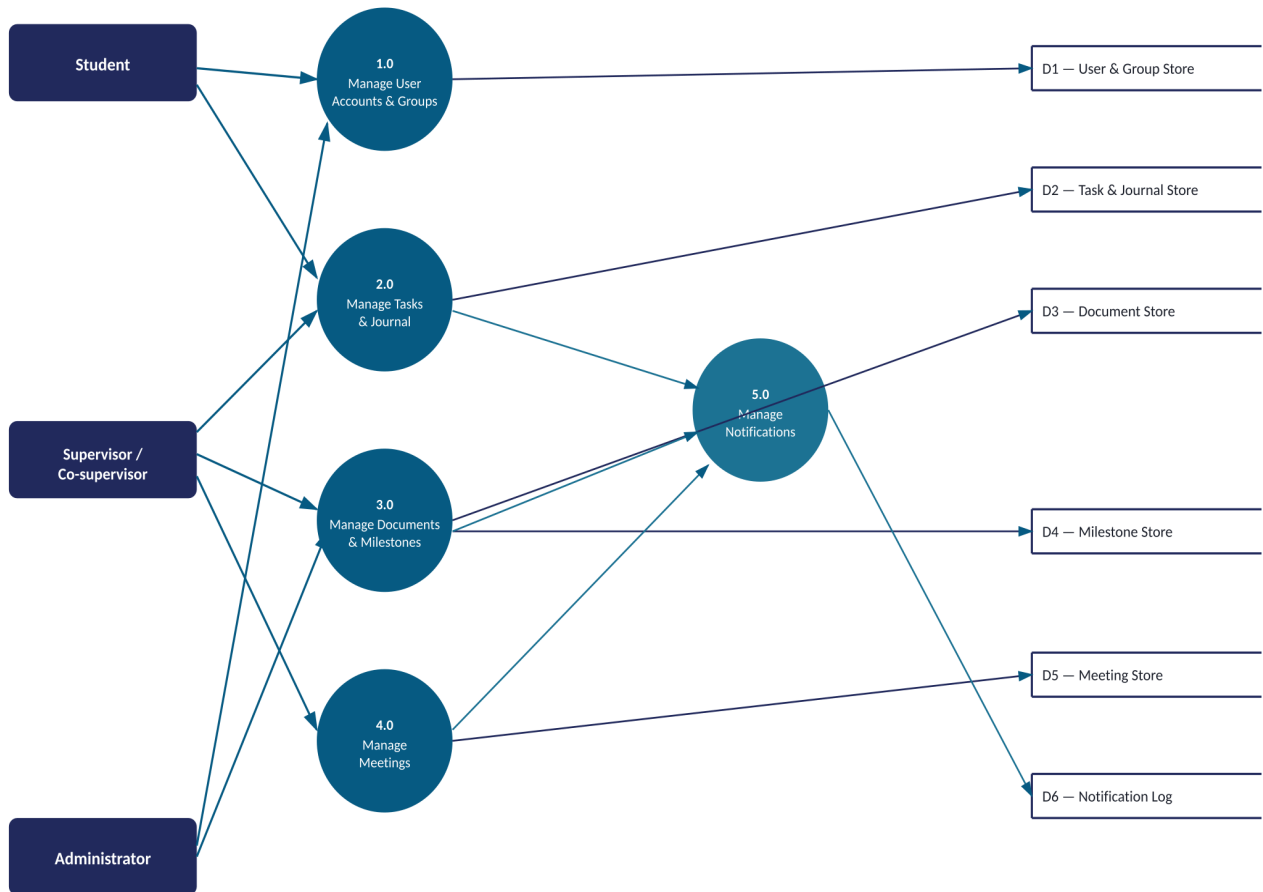


Figure A.2 - ThesisSync Data Flow Diagram (Level 1)

A.3. Process Flow Diagram — Document Submission and Review

This process flow diagram traces a single document submission from upload through supervisor review to milestone approval, including the revision loop when a submission needs further work.

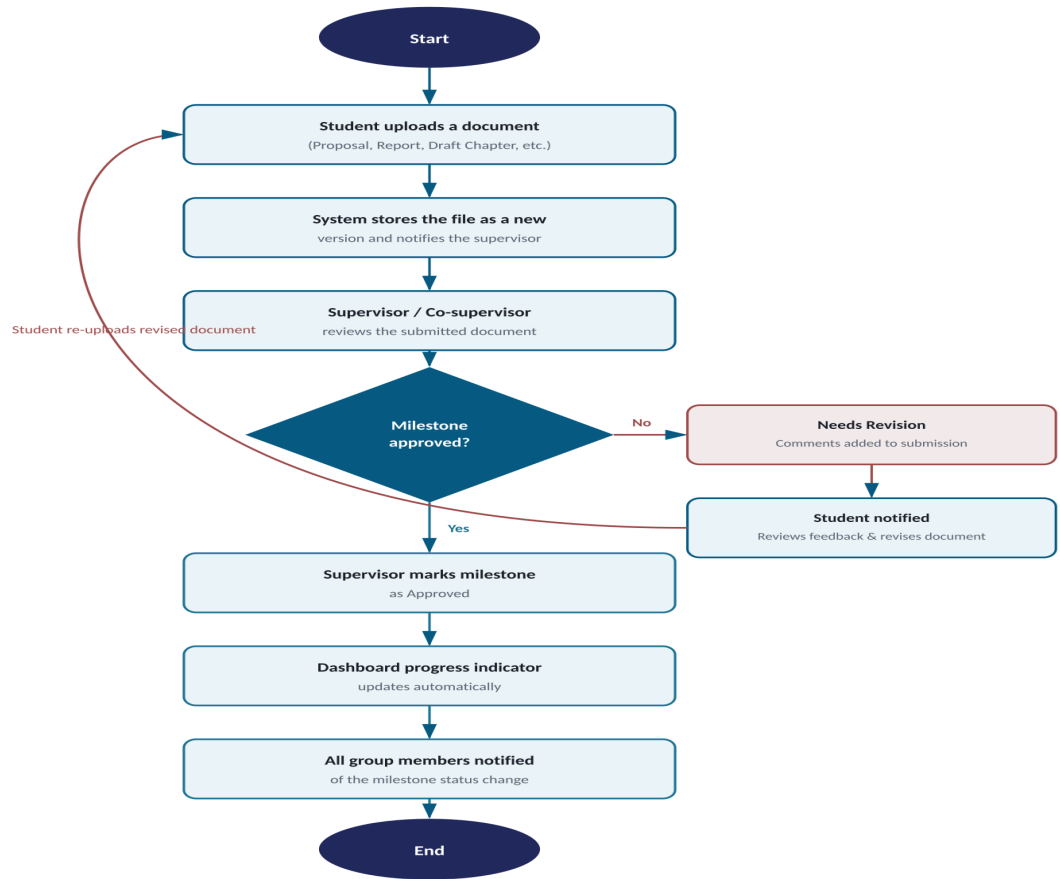


Figure A.3 – Document Submission & Review Process Flow

A.4. Entity Relationship Diagram

The entity relationship diagram models the core data entities of the platform - User, ThesisGroup, Task, JournalEntry, Document, DocumentVersion, Comment, Milestone, Meeting, and Notification - and the relationships that connect them.

A.5. Use Case Diagram

The use case diagram summarizes the key interactions each actor — Student, Supervisor, Co-supervisor, and Administrator — has with the system, from registration and task management through to review, approval, and administrative reporting.

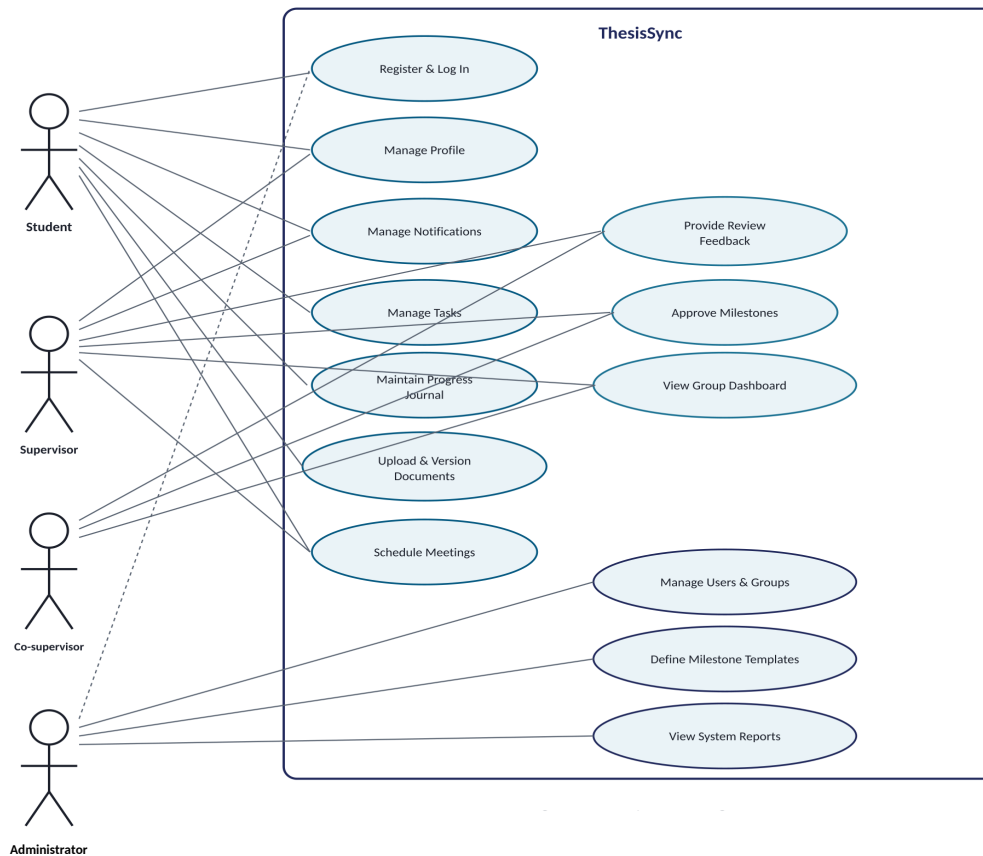


Figure A.5 - ThesisSync Use Case Diagram

6.5. Appendix B: Issues List

Open issues and questions to be resolved during development:

- Final decision on the hosting platform for deployment (university infrastructure vs. third-party cloud provider)
- Final choice of backend framework and database engine
- Storage quota per thesis group for the file repository
- Whether an external-evaluator (defense committee) access role is needed for this phase or deferred to a later one

- Final list of milestone templates to be applied uniformly across thesis groups

[Note: This list will be updated throughout the project lifecycle.]